# Forecasting product quality using peephole long short-term memory


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# Table of content

- Introduction
- Proposed model
- Forecasting Scheme
- Experiment Result
- Conclusion

# Introduction

# Introduction

- Forecasting is essential in manufacturing to **predict the future health of manufacturing systems and plan maintenance activities**.
- Existing studies focus on using process data (**sound, vibration, magnetic**) for predicting system health.

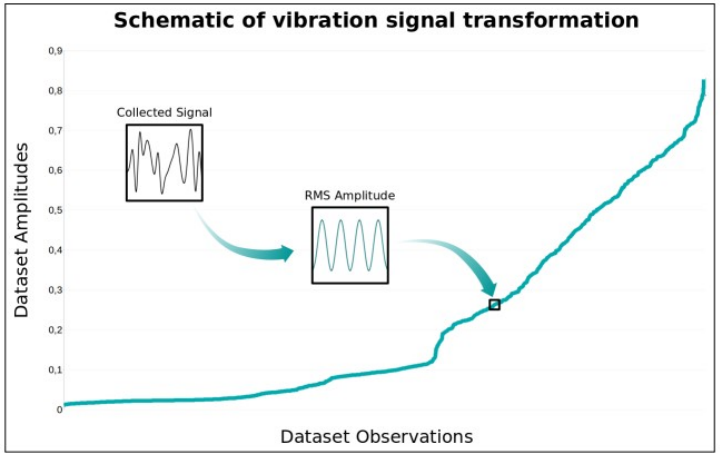



**Figure 1:** An example of the correlation between vibration data and system health (represent by the increase in vibration amplitudes)

- Limited studies use **product quality data** for forecasting in manufacturing.
- Analyzing product quality data can help identify **patterns, trends, and potential issues in manufacturing processes**.

# Challenges in Product Quality Forecasting

Product quality forecasting has several challenges as follows:

- Short-term product quality time series is **nonlinear**, **non-stationary**, and **non-seasonal**.
- **Limited additional information** is available for learning in univariate forecasting compared to multivariate forecasting.
- Forecasting accurately over **a long-time horizon (multi-step ahead)** requires effective methods.

In this study, we propose Peephole Long Short-Term Memory (LSTM)-based Recurrent Neural Networks (RNNs).

- This model can **effectively** forecast product quality data.
- Experiments demonstrate **the superiority** of LSTM-based RNNs over other forecasting methods.

# Proposed Model

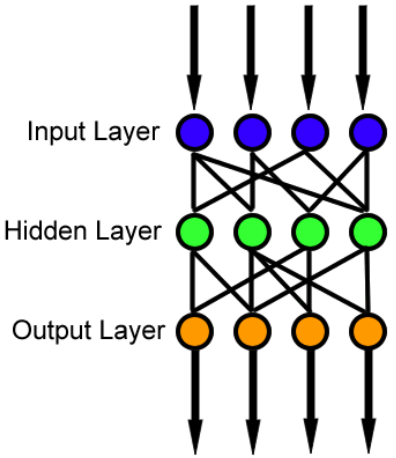
# Feedforward Neural Networks (FNNs)

**Feedforward Neural Networks (FNNs):** is a neural network architecture in which information goes forward from the input to the output layer.

⇒ **However, when the size of the input increase, we have to change the number of input units.**

**Recurrent neural networks (RNNs)** have the same elements as FNNs but have additional recurrent connections.

⇒ **These connections help RNN has the flexibility to capture the information of sequence data with different length.**



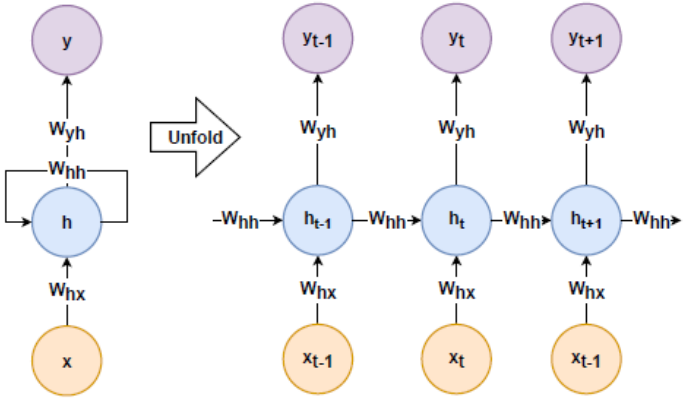




Figure 2: The architecture of FNNs and RNNs

# Long-Short-Term Memory (LSTM)

When the length of the input is relatively large, RNNs have a problem with optimization (**gradient vanishing**)

To overcome the vanishing problem, they use the Long-Short-Term Memory (LSTM) cell in RNNs

- The LSTM has additional forget, input, update, and output gates.

- These gates can learn to **memorize long-term and short-term information separably and effectively.**

**In order to get better memory utilization, in this study, the peephole connections were added to LSTM.**

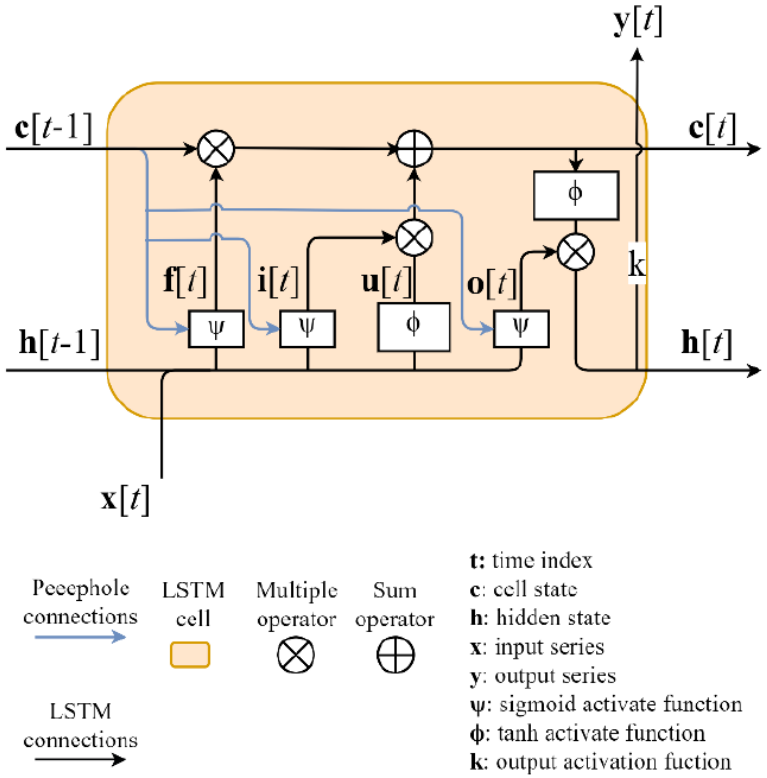



Figure 3: The architecture of peephole LSTM cell

# Forecasting Scheme

# Forecasting Scheme

To forecasting multiple ahead-of-quality data, we use a “Multi Input Multi Output (MIMO)” strategy

Specifically,

- Given the input time series **s(t) = {$s_1$, $s_2$, …, $s_{nT}$}** with $n_T$ is input length
- LSTM transforms **s(t)** into output time sequences:
  - **Output: y = {$y_1$,$y_2$,…,$y_H$}** with H is the output length

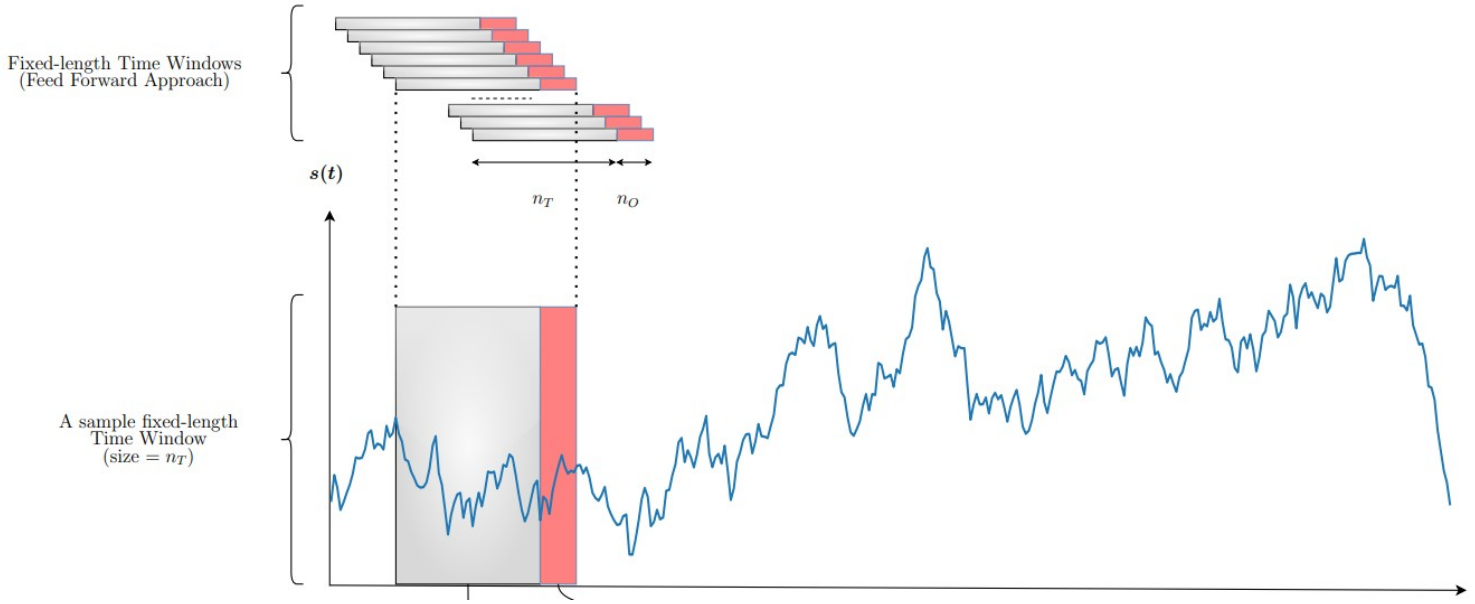



Figure 4: Forecasting scheme

# Experiment Result

# Experimental Results

We evaluate the model's performance based on 3 metrics:

- root mean square error (RMSE)
- mean absolute percentage error (MAPE),
- and R2 score

Given the forecasting result $\hat{\mathbf{y}} = \{\hat{y}_1, \hat{y}_2, \ldots, \hat{y}_H\}$ and the actual value $\mathbf{y} = \{y_1, y_2, \ldots, y_H\}$

$$RMSE = \sqrt{\frac{1}{H}\sum_{1}^{H}(y_i - \hat{y}_i)^2}$$

$$MAPE = \frac{100\%}{H}\sum_{1}^{H}\left|\frac{y_i - \hat{y}_i}{y_i}\right|$$

$$R^2 = 1 - \frac{\sum(y_i - \hat{y}_i)^2}{\sum(y_i - \bar{y})^2},$$

We compare **peephole LSTM** with 3 baseline methods:

- Support vector regression
- SARIMA
- Convolutional neural network

# Forecasting in Airline Passenger dataset

To demonstrate the generalize, we apply our model to a public dataset “Airline Passenger”. We use a forecasting horizon equal to 12 (H = 12)

**Table 1. Forecasting RMSE, MAPE, and R2 score for the Airline Passenger Data Set**

| Forecasting method | RMSE | MAPE | R2 |
|---|---|---|---|
| SVR | 0.1565 | 0.1749 | 0.0970 |
| SARIMA | 0.1393 | 0.1703 | 0.0499 |
| CNN | 0.1659 | 0.1921 | **0.5184** |
| LSTM | **0.1059** | **0.1509** | 0.4508 |

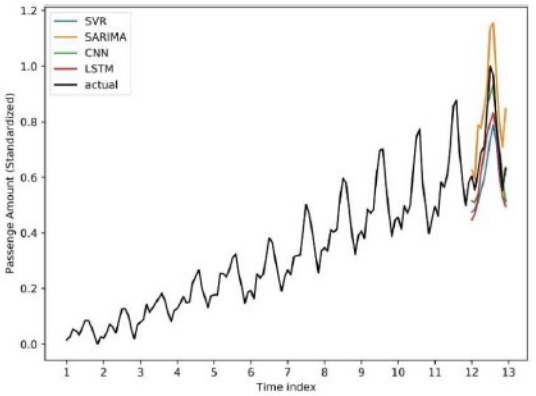



**Figure 3. Comparison of the results for the international airline passenger data set both within and outside of the sample**

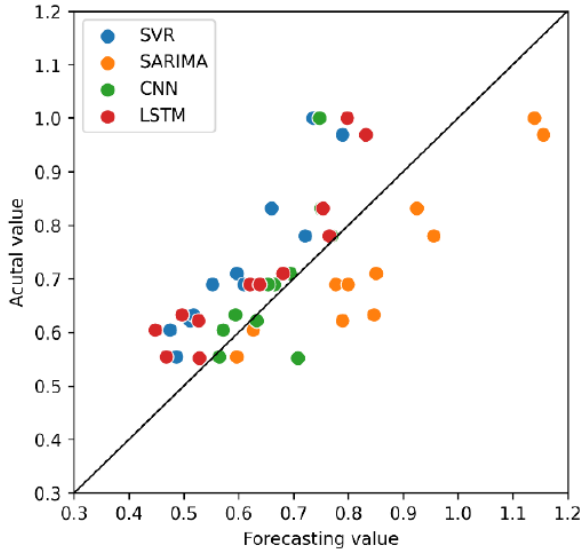



**Figure 5. Predicted and actual value of the forecasting methods for international airline passenger data set.**

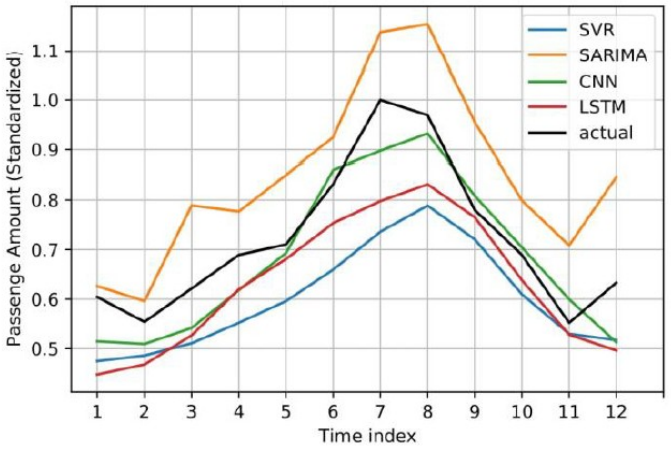



**Figure 4. Comparison of forecasting performance for the international airline passenger data set.**

# Forecasting in Quality Dataset

We utilized 720 observations of product quality data to predict future quality over a 24-step horizon (H = 24). The result is shown as flowing:

**Table 2. RMSE, MAPE, and R2 score results for the Product Quality Data Set**

| Forecasting method | RMSE | MAPE | R2 |
|---|---|---|---|
| SVR | 0.1660 | 0.2220 | 0.2650 |
| SARIMA | 0.1918 | 0.3550 | 0.1227 |
| CNN | 0.1509 | **0.1820** | **0.4640** |
| LSTM | **0.1280** | 0.2320 | 0.2261 |

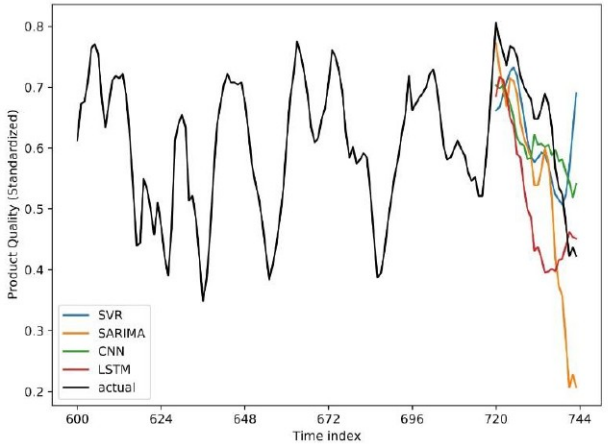



**Figure 6. Comparison of in-sample and out-of-sample performance for the product quality dataset**

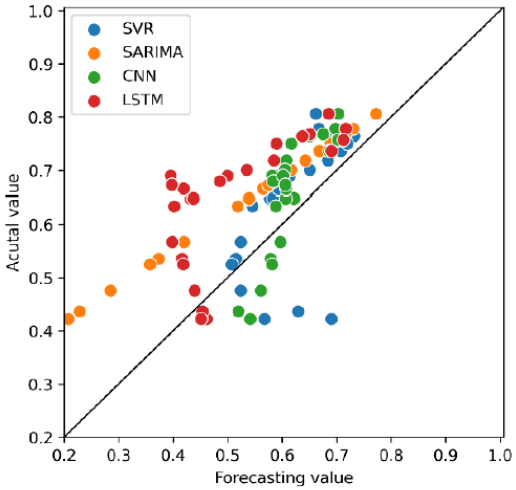



**Figure 8. Predicted and actual value of the forecasting methods for international airline passenger data set.**

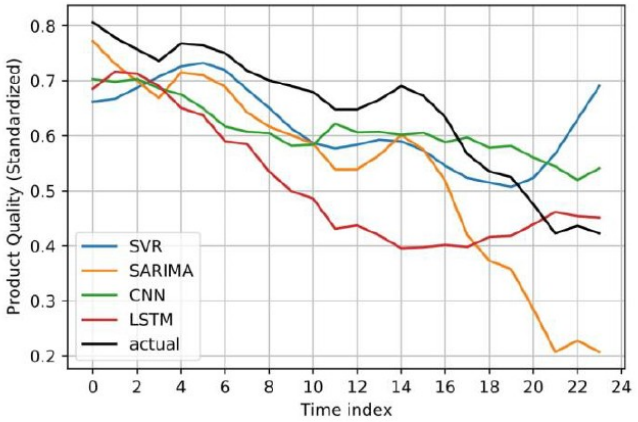



**Figure 7. Comparison of forecasting results for the product quality data set.**

# Conclusion

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In summary, this study recommends using **LSTM-based RNN for accurate short-term product quality forecasting** by leveraging long-term dependencies in time series data.

The proposed method **outperforms traditional forecasting techniques** in predicting complex and non-stationary product quality patterns, as demonstrated through tests on **benchmark airline passenger data** and **a comprehensive roasting process dataset**.

Questions and Discussion